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MILITARY SPECIFICATION

MACHINING CENTERS, VERTICAL, SINGLE SPINDLE, COMPUTER NUMERICAL CONTROL (CNC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers computer numerically controlled (CNC), single vertical spindle type machining centers, with an automatic tool changer and other components or equipment as described herein.

1.2 Classification. Machining centers covered by this specification have three CNC controlled axes and are of the following sizes and styles. Also covered is an additional controlled axis rotary table which may be optionally specified (see 3.4.10.1.1). The size and style machining center to be supplied shall be as specified (see 6.2.1).

	Spindle motor horsepower	Mill capacity Cubic inches/minute (SAE 1020 Steel)	CNC controlled travel x-y-z, inches
Size 1	5	5	40-20-8
Size 2	7	7	40-20-10
Size 3	7	7	50-26-10
Size 4	10	10	40-20-12
Size 5	10	10	50-26-12
Size 6	10	10	80-26-12
Size 7	15	15	54-28-14
Size 8	20	20	54-28-24

Style A - Tool Magazine capacity of 15 or more tools
Style B - Tool Magazine capacity of 20 or more tools
Style C - Tool Magazine capacity of 30 or more tools
Style D - Tool Magazine capacity of 40 or more tools
Style E - Tool Magazine capacity of 60 or more tools

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in the that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.
ANSI B5.50 - "V" Flange Tool Shanks for Machining Centers with Automatic Tool Changers.

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(Applications for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA) STANDARDS

NFPA No. 79 - Electrical Standards for Metalworking Machine Tools.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA) STANDARDS

RS-232 - Interface between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Interchange

RS-244 - Character Code for Numerical Machine Tool Control Perforated Tape

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machine Tools

RS-274 - Interchangeable Variable Block Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines

RS-281 - Construction Standards - Numerical Machine Tool Control

RS-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape

RS-447 - Operational Command and Data Format for Numerically Controlled Machines

RS-449 - General Purpose 37-Position and 9-Position Interface for Data Terminal Equipment and Data Circuit Terminating Equipment Employing Serial Binary Data Interchange.

RS-494 - 32 Bit Binary CL Exchange (BCL) Input Format for Numerically Controlled Machines

(Applications for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, DC 20006.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA) STANDARDS

MG-1 - Motors and Generators

(Applications for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial Packaging, Standard Practice for.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete machining center for first article inspection and approval (see 4.2 and 6.4).

3.2 Design. The machining center shall be new and one of the manufacturer's current models. Its design shall include features and components necessary for maintaining alignment and accomplishing drilling, tapping, milling, and boring performance required herein. The machine shall have a vertical axis spindle assembly housed within a machining head which shall be supported by a column and floor mounting base. The base shall also support a saddle and/or worktable which shall be arranged to support and secure the workpiece for machining operations. The worktable and/or column shall have slide motion (travel) as necessary for altering the relationship of the spindle with the workpiece in the x and y axes. Unless otherwise specified (see 6.2.1), z axis movement shall be accomplished by either a machining head or machining head and quill, sliding vertically along the column ways to and from the worktable or with the machining head affixed to the column and by a quill sliding within the head and to and from the worktable. In the latter case, the worktable shall also be vertically adjustable. Tools, toolholders and adapters shall be inserted and removed from the spindle by a tool changer with storage magazine. Tool change, spindle operation, and motion in the x, y, and z axes shall be directed by a computer numerical control (CNC) system. The machine and its numerical control system shall constitute a completely functional system with built-in operations programmed by coded punch tape and/or other devices. The machine functions shall be controlled by manually operated control devices, semi-automatically by manual data input devices, and automatically from tape input and part programs in data memory. All parts of the machine or system that are subject to wear, breakage, or distortion shall be accessible for adjustment, replacement and repair.

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3.2.1 Operating controls. Operating controls for the machining center shall be arranged on the CNC console or machine pendant. Controls such as push-buttons, dials, and switches shall be functionally arranged and clearly identified to facilitate use.

3.2.2 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.3 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with means for lubrication. Oil reservoirs shall be fitted with oil level sight gages. Manually operated systems for ways and remote areas shall have control handles mounted in accessible locations. All oil holes, grease fittings, and filler caps shall be easily accessible.

3.2.4 Interchangeability. To provide for replacement of worn parts or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.2.5 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.3 Construction. The machine shall be so constructed that when installed and connected to power, it will be ready for operation upon filling with operating fluids. The machine shall be constructed of parts which are without defects and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

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3.3.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed in such a manner as to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings and welded parts shall be cleaned and free from sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's standard commercial practice.

3.3.5 Painting. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with manufacturer's standard commercial practice which shall be not less than one coat of primer and one coat of finish color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. Unless otherwise specified (see 6.2.1), feed screw dials shall be graduated in either the inch or metric (SI unit) system. Graduations to indicate stock removal or tool movement shall be in increments of not more than 0.001 inch or the equivalent on the metric scale. The size of dial faces shall be such that the graduations are easily read. Dial faces shall be permanently and legibly engraved, stamped, or etched, and shall have a non-glare finish.

3.3.8 Plates. All words on speed and feed indicating plates and on instruction plates shall be in the English language, engraved, etched, embossed or stamped in boldface characters on a contrasting background.

3.3.9 Gears. All gears shall be constructed from a material suitable for the intended purpose. All metal gears shall be machined and heat treated by a process that will impart the hardness, surface finish and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure or premature deterioration and wear. All gears shall be manufactured to definite established standards to provide for interchangeability and parts replacement.

3.4 Components. The machining center shall include components consisting of a base, column, saddle, worktable, machining head, spindle assembly, tool changer, coolant system, electrical system, and numerical control system.

3.4.1 Base. The base shall be a casting of iron or iron alloy or a fabricated steel weldment, ribbed and braced to minimize distortion and deflection. The base shall form a foundation for the entire machine and be

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integrally constructed with the column or be separately constructed with the column machined, securely attached, and in either case, be sufficiently rigid to maintain mutual component alignment for assuring accuracy required herein. The base shall have guideways that are proportioned to fully support the saddle and/or column and worktable throughout its full range of travel. The base shall have means for leveling the machine on a flat mounting surface. Hydraulic or coolant fluid reservoirs may be housed within the base or be supplied as separate external units. Reservoirs shall have means for indicating fluid levels. The base shall have access or removable plates as necessary for servicing components housed within.

3.4.2 Column. The column shall be a casting of iron or iron alloy or a fabricated steel weldment of box-like construction, internally ribbed and braced to minimize distortion and deflection under normal tool loads. The column shall be designed to support and accommodate the machining head and have guideways to facilitate vertical movement of the head or worktable. If column movement is used to provide either X or Y axis movement, the column shall be precision fitted to the guideways of the base.

3.4.3 Saddle. The saddle slide shall be a casting of iron or iron alloy or a fabricated steel weldment, rigidly constructed and precision fitted to the guideways of the base. The saddle shall also have guideways for supporting and traversing the worktable throughout its full range of travel. Guideways of the saddle and all other components of the machine shall be designed with friction characteristics that will permit precise control of slide movement without sticking or overshooting. When specified (see 6.2.1), the guideways of the machines shall be hardened and ground having a hardness not less that Rockwell C-50. Mating ways shall have means for holding them in alignment to prevent cutting forces from separating slide surfaces or distorting feed screws. Means such as servo locks shall be provided that automatically secure slide surfaces in place immediately upon completing numerical control directed moves. The guideways shall be shielded from chips, coolant fluid and other contaminants by telescoping metal covers or similar means.

3.4.4 Worktable. The worktable shall be a casting of iron or iron alloy, internally ribbed and proportioned to support and withstand maximum capacity workpiece weights and tool thrust loads. The worktable shall be precision fitted to the ways of the saddle and be the compound type, arranged to provide workpiece travel in x and y axes. The top of the worktable shall be machine finished for a work mounting surface. The worktable shall have a coolant trough or similar means for draining spent coolant into its reservoir. Unless otherwise specified (see 6.2.1), the number of T-slots shall be the manufacturer's standard. The size and configuration of T-slots shall be in accordance with ANSI B5.1. When required, the worktable shall be furnished with a subplate of the characteristics specified (see 6.2.1).

3.4.5 Machining head. The machining head shall be a casting of iron or iron alloy or a fabricated steel weldment designed and proportioned to

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house and support the quill (as required), spindle assembly, and spindle rotational drive with motor and related components necessary for accomplishing drilling, tapping, milling, and boring. The machining head shall hold the spindle in axial alignment perpendicular to the work mounting surface of the worktable and shall be designed to prevent distortion of the head due to torsional forces of the drives. The z axis movement shall be servo driven and be fully controlled by the numerical control system. The machine shall have stops that are located at the upper and lower extremes of Z axis travel to prevent excessive overtravel.

3.4.5.1 Spindle assembly. The spindle shall be of alloy steel hardened and ground to a finish not rougher than 32 microinches. The spindle shall be supported and held in axial and radial alignment by ball or roller bearings. The spindle shall be mounted in the machining head with its bearings designed and arranged to compensate for thermal expansion. The spindle shall have positive means for accepting, retaining and releasing tools and tool holders inserted by either the automatic tool magazine or the operator. Unless otherwise specified (see 6.2.1), nose, bores, and other characteristics of the spindle shall be in accordance with ANSI B5.50. The spindle shall be equipped to orient the automatically inserted tools to maintain the same radial position they occupied in the tool magazine. The machine shall drive the spindle in both the clockwise and counterclockwise directions of rotation for accomplishing the full range of spindle speeds specified herein. The machine shall be arranged to automatically stop spindle rotation for tool magazine. The machine shall have overload protection which automatically stops the movement of all slides and the rotation of the spindle should it stall from overload.

3.4.6 Tool changer. The machine shall have an automatic tool changer with a magazine that is capable of storing not less than the tool capacity required for the style machine specified in the contract or order. The magazine shall be of the rotary drum design or of any other closed loop recirculating configuration with the capability of presenting a preprogrammed tool to the tool changer for exchange with tool held in the spindle. Tool change shall be accomplished by means of a power-actuated mechanism which shall either interchange the tool in the spindle with a tool selected from the tool magazine or remove the tool from the spindle then insert the tool from the magazine. The magazine shall permit convenient loading and unloading of tools, tool holders, and adapters during operational set-up. The tool changer shall be arranged for random tool selection which shall include the capability of reusing a tool that has been returned to the magazine after a program operation.

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3.4.7 Coolant system. Unless otherwise specified (see 6.2.1), the machine shall be supplied with a complete flood and mist type coolant system with numerically controlled selection of coolant type and on/off of coolant flow. The system shall have adequate capacity for all types of machining operations for which the machine is designed and shall have means for regulating the flow of flood coolant and for adjusting mist coolant supply as required. All components necessary to make the system completely functional shall be furnished including reservoirs, chip screens, filters, drain troughs, and clean-out plates.

3.4.8 Hydraulic system. The hydraulic system, when supplied, shall be complete with pumps, valves, piping, cylinders or motors, reservoirs and pressure controls. Overpressure protection shall be provided on the high pressure line from the hydraulic pump to prevent damage to components. The system shall include filters to insure delivery of clean fluid. The reservoir shall have a sight gage to indicate the fluid level. The fluid temperature shall be maintained within the proper operating range during all normal operations of the machine. When required, noise level and noise suppression enclosures for stand-alone hydraulic units shall be as specified (see 6.2.1).

3.4.9 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to NFPA No. 79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 230 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA-79 for the disconnect fuse rating.

3.4.9.1 Motors and motor accessories. Motors shall have ball or roller bearings and shall be 60-minute duty rated. Motors shall be either AC 3 phase, 60 Hz, or industrial DC motors conforming to NEMA Standard MG1 and equipped with solid state rectifiers, or shall be stepping type motors with not less than 50 incremental steps per revolution. Unless otherwise specified (see 6.2.1), each electric motor shall meet the requirements of a dripproof enclosure.

3.4.10 Numerical control. The machining center shall have a fully automatic solid state soft-wired, integrated circuit type numerical control. The machine control unit (MCU) shall provide automatic control of machine functions, operating modes, three linear axes movement, spindle operation, tool change, and other part program directed functions for machining centers of the type specified herein. Controlled axes shall be identified in accordance with EIA Standard RS-267. The control shall have linear and circular interpolation and be capable of simultaneous control of three linear axes. Control features shall include a programmable interface, buffer storage, fixed cycles, part program storage, part program and buffer edit capability, and control diagnostics. The MCU shall direct machine

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functions from part program data stored in memory, manually input or directly from the input media specified herein. The control shall initiate a halt or an error signal should a fault condition occur in the control. All necessary executive program routines shall be furnished in the form of 1-inch 8-channel binary coded decimal punched tape or 8-inch flexible disk for controls having volatile memory. The control unit shall conform to EIA Standards RS-281. A line voltage of ±10% from normal shall not adversely affect the numerical control system function. The control shall be capable of functioning in ambient temperatures ranging from 50 to 120 degrees F and humidity ranging from 5 percent to 95 percent noncondensing. The control shall automatically shut down when internal operating temperature exceeds safe operating temperature. Control resolution shall be 0.0001 inch or 0.001mm for each linear axis and not more than 0.001 of a degree for each rotary axis. When binary cutter location (BCL) data is specified (see 6.2.1) the numerical control shall have the necessary hardware and software to perform all geometry dependent functions which are ordinarily performed by the postprocessor.

3.4.10.1. MCU. The MCU shall provide full contouring control of the machines three linear axes and positioning control of the rotary table for performing drilling, tapping, boring, reaming, and contour milling operations by simultaneous movement of each axis.

3.4.10.1.1 Rotary table, a axis. When specified (see 6.2.1), the control system shall also be configured for an a axis and the machine shall be furnished with a rotary table. The a axis control shall direct rotation of the rotary table to any of the program selectable (degree) positions specified (see 6.2.1). The rotary table shall be of the contouring type or positioning type as specified (see 6.2.1). The rotary table shall be either an integral component of the machine or an attachment to the worktable. The top of the rotary table shall have T-slots for workpiece attachment. When specified (see 6.2.1), the rotary table shall be constructed so that the workpiece shall rotate about the X axis.

3.4.10.2 Data input.

3.4.10.2.1 Punched tape data input. Unless otherwise specified (see 6.2.1) the input media for the control shall be 1-inch, 8-channel punched tape. The data input format and codes shall comply with EIA Standards RS-274, RS-358 and RS-447 or when specified (see 6.2.1) RS-244 shall apply in lieu of RS-358. The control shall be equipped with a photoelectric tape reader capable of holding, feeding and reading tape prepared in the format specified at a rate not less than 150 characters per second. Unless otherwise specified (see 6.2.1), the tape reader shall be supplied with standard feed and take-up spools 5-1/4 inches in diameter. The tape reader shall be housed in a dust free area of the numerical control cabinet and shall be easily accessible for cleaning and tape handling. The control shall have a tape error detection capability to continuously check for the following errors:

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a. Odd parity per EIA RS-244
b. Even parity per EIA RS-358
c. Misaligned sprocket holes
d. No sprocket holes

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display screen to indicate a fault condition.

3.4.10.2.2 Flexible diskette data input. When 32 Bit Binary Cutter Location (BCL) Exchange Input for Numerically Controlled Machines is specified (see 6.2.1), the input media shall be an eight inch flexible diskette recorded in accordance with 3.4.10.2.2.1. The input data format shall comply with EIA Standard RS-494. Functions 10.1, 10.2 and 10.3 of Section 10, EIA-RS-494 shall apply. Manufacturers providing functions or features that cannot be addressed by command codes currently covered by EIA-RS-494 shall apply to the Electronic Industries Association, IE-31 Committee on Numerical Control Systems and Equipment for an extension of the standard codes to cover the new features or functions. The control shall be equipped with an eight inch flexible diskette reader. The reader shall have the capability to perform a cyclic redundancy check of the input data and be capable of reading not less than 15,000 characters per second.

3.4.10.2.2.1 Flexible diskette format. When an eight inch flexible disk is specified as the input media to the numerical control it shall be a single density recording on one side using a soft sectored format. The disk shall be divided into 77 distinct locations on the single side disk. Tracks shall be assigned numbers from 00 through 76. Track 00 shall be nearest to the edge of the disk and track 76 nearest the center. Tracks 00, 74, 75 and 76 shall not be used. There is an index hole in the jacket and the disk that shall be used for timing functions when a beam of light passes through it. Recording shall be accomplished by rotating the disk inside its jacket exposing the disk at the read/write head slot cut in the jacket. Data shall be recorded only on the side of the disk that is exposed to the read/write head. Each track shall be divided into 26 addressable sectors which contain 128 bits. All sectors shall be identified by the file starting sector number with track 00, sector 1, designated as 0. The first part program shall be located at sector 1E hexidecimal corresponding to track 1, sector 5. Where two byte numbers are referenced the high order byte shall be recorded first and the low order last. The data stored in a sector is called a record. Track 1, sectors 1 through 4 shall contain the program directory of up to 16 part program names in ASCII. Each file shall be packed on the disk in the same order as its directory entry. Unused directory entries shall point to the next available sector after the last file and contain the 30 character ASCII space characters. The last sector of each file shall be padded with end of file (EOF) records (8000 0000). Diskette addressing shall comply with the requirements of table IV.

3.4.10.2.3 Manual data input. A means for manual data input (MDI) shall be furnished to permit manual entry of alphabetical, numerical, and special

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characters necessary for control of all programmable machine functions. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

3.4.10.3 Data display. A cathode ray tube (CRT) capable of displaying at least 250 characters simultaneously, shall be provided. Unless otherwise specified (see 6.2.1), the display shall, as a minimum, display the following data:

a. Four digit block sequence number.
b. Active block of stored data.
c. Actual position of each axis.
d. Feedrate and override condition.
e. Control and system error messages.

3.4.10.4 Control system modes of operation. The operational modes of the NC control shall include manual, automatic, single block, MDI, sequence number search, and optional stop.

3.4.10.5 Departure control. The maximum departure permitted by a single command in either incremental or absolute mode shall be not less than plus or minus 99.9999 inches or 999.999mm. The control shall be capable of automatically controlling the acceleration and deceleration of the slides. The resolution of slide movement in either direction shall not be more than 0.0001 inch or 0.001mm.

3.4.10.6 Absolute/incremental input. The control shall have absolute/incremental input capability. This mode shall be selected by a preparatory "G" code which may be switched during the tape program. Not required if BCL input is specified (see 6.2.1).

3.4.10.7 Automatic tape code recognition. The control shall have the capability to automatically recognize and read tape punched with either EIA RS-244 or RS-358 (ASCII) character codes. Not required if BCL data input is specified (see 6.2.1).

3.4.10.8 Switchable inch/metric input. The control system shall provide switchable inch/metric input capability.

3.4.10.9 Part program storage. Unless otherwise specified (see 6.2.1) part program storage shall be in either solid state or bubble type memory. The MCU shall be capable of storing in memory the content of not less than 250 feet of postprocessed part program tape data unless otherwise specified (see 6.2.1). Not applicable if BCL data input is specified (see 6.2.1).

3.4.10.10 Standby battery power. Unless otherwise specified (see 6.2.1), standby battery power shall be provided for controls having

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volatile memory, to hold the memory provided at full load for a period of not less than 70 hours when the primary power source fails.

3.4.10.11 Part program edit. A program edit capability shall be provided to allow block insertion, deletion or modification of any part program in the part program memory. Data related to part program edit shall be input by the manual data input provided on the control panel. Edited program data entered into program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

3.4.10.12 Feed rate override. A feed rate override control shall be provided to allow manual adjustment of the programmed feed rate from zero to not less than 120 percent of the programmed rate in either continuous adjustment or in steps of not more than 10 percent. The feed rate override control shall have a position that allows return of feed rate to that programmed.

3.4.10.13 Spindle speed override. A spindle speed override shall be provided to allow manual adjustment of spindle speed in either continuous adjustment or in steps of not greater than 10 percent each. The spindle speed override control shall control not less than the range of override percentages between 60% and 120% of programmed speed. The spindle speed override control shall have a position that allows return of speed to that programmed.

3.4.10.14 Sequence number search. The control shall have the capability to permit the operator to search for any desired sequence number.

3.4.10.15 Mirror image. The machine shall be capable of producing either right or left-hand from the same tape data through the inversion of x and y axes. Inversion of each axis shall be by keyboard or switch entry. Mirror image shall not affect the directional sense of manual controls. Reverse settings shall be displayed on the data display.

3.4.10.16 Preparatory functions (G-Codes). The preparatory function codes of the control shall be in accordance with EIA Standard RS-274. If the BCL data input is specified, codes in accordance with EIA-RS-494 shall be applied.

3.4.10.17 Miscellaneous functions (M-Codes). The miscellaneous function codes of the control shall be in accordance with EIA Standard RS-274. If the BCL data input is specified, codes in accordance with EIA-RS-494 shall be applied.

3.4.10.18 Block delete. The control shall have a block delete feature to permit by-passing of blocks of programmed data.

3.4.10.19 Buffer storage. The control shall have buffer storage for transferring command data from the tape reader to internal storage without delaying the next incoming command and without interrupting the machine

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functions. The internal storage shall delay or store multiple blocks of data until required by the controlling devices.

3.4.10.20 Block-by-block read. The control shall have a block-by-block read function to provide for reading information one block at a time.

3.4.10.21 Tool length offsets. The control shall have a tool length offset feature to permit adjustments to the programmed tool length by means of manual data input. The tool length offset feature shall have a range of not less than the following:

a. Inch system ±0.0002" to ±1.0000"
b. Metric system ±0.002mm to ±25.400mm.

3.4.10.22 Cutter compensation. Cutter compensation shall be provided to compensate for cutter diameters differing from those utilized in programming. The cutter compensation shall have a range of not less than the following:

a. Inch system ±0.0002" to ±1.0000"
b. Metric system ±0.002mm to ±25.400mm.

3.4.10.23 Fixture compensation. A fixture compensation feature shall be provided to permit compensation for fixture position errors.

3.4.10.24 Reversal error compensation. Reversal error compensation shall be provided for each axis of motion to compensate for drive gear backlash when slide reversal occurs. A means shall be provided to input new compensation values when required.

3.4.10.25 Axis calibration. Error compensation shall be provided to correct repeatable errors determined to exist due to leadscrew and feedback device inaccuracies. A means shall be provided to input new error compensation values when required.

3.4.10.26 Additional features. Unless otherwise specified (see 6.2.1), the following additional numerical control features shall be provided:

a. Four quadrant programming
b. Incremental and continuous jog for each axis
c. Leading zero suppression or decimal point programming
d. Inches per minute (IPM)/Millimeters Per Minute (MMPM), programming in all modes of operation
e. Tape edit keylock.

3.4.10.27 Operator control panel. Unless otherwise specified (see 6.2.1), the control unit shall have a control panel which provides at least the following control functions: Emergency Stop; Cycle Start; Optional Stop; Mode Selection; Feed Hold; Feedrate Override; Spindle Speed Override; Jog; Jog Direction; and EOB Stop.

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3.4.10.27.1 Pendant control. When specified (see 6.2.1) a pendant control shall be furnished as an operator control panel. The pendant control shall be connected to the machine to enable the operator to control machine functions while in a close observation position. Unless otherwise specified (see 6.2.1), the pendant control shall be capable of the same functions as stated for the operator control panel in paragraph 3.4.10.27.

3.4.10.28 Reference zero. The control unit shall have a reference zero to move each axis to a reference limit switch to establish synchronization and reset all registers to zero.

3.4.10.29 Set zero. The control unit shall have a "set zero" feature to permit the specified axis register to be reset to zero establishing present axis location as zero.

3.4.10.30 Program tryout. The control shall have a program tape tryout feature to permit new part programs to be read through the control without machine motions.

3.4.10.31 Operational software. Operational software shall be resident in the control and shall contain all logic necessary to effectively operate the machine tool. The software shall provide for meaningful diagnostics relating to the utilization of the controls, machine tool, part program, and operator's actions.

3.4.10.31.1 Programmable interface. A software program for accommodating the softwired interface between the numerical control and machine shall be provided as part of the executive program or as a separate program in the form of one inch, eight channel punched tape complying with EIA-RS-244 or eight inch flexible diskette as appropriate for the input media specified (see 6.2.1).

3.4.10.32 Maintenance diagnostics system. Maintenance diagnostics software shall be furnished in either the control as non-volatile memory or in the form of one inch, eight channel punched tape complying with EIA-RS-244 or eight inch flexible disk as appropriate for the input media specified (see 6.2.1) to provide for diagnostic checking of the machine control unit. The diagnostic software shall test, exercise, and display failures, at least to board level.

3.4.10.33 Peripheral equipment interface. Unless otherwise specified (see 6.2.1), the control shall be equipped with an interface complying with RS-232 and RS-449.

3.4.10.34 Axis jog. The control panel or pendant shall have manual controls for accomplishing both single and continuous, low and high speed jog movements of the slides for each controlled axis in both the plus and minus direction.

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3.4.10.35 Auxiliary functions. Unless otherwise specified (see 6.2.1), the CNC shall respond to the following program commanded auxiliary functions: interpolation, fixed cycles, and dwell. Unless otherwise specified (see 6.2.1), miscellaneous functions shall include program stop, optional program stop, end of program, spindle CW, spindle CCW, spindle off, coolant on-off, tool change, end of program (rewind), and program re-start. System preparatory and miscellaneous functions shall be coded in accordance with EIA RS-274 and EIA RS-447.

3.4.10.36 Canned cycles. The following canned cycles shall be provided:

a. Drilling
b. Boring
c. Tapping

3.4.10.36.1 Macro program cycles. The following flexible macro program cycles shall be provided:

a. Milling
b. Drilling
c. Boring
d. Tapping

3.4.10.37 Tape punch unit. When specified (see 6.2.1), the system shall have a tape punch unit for producing punched tapes from edited programs of tape being input and for programs stored in data memory. The unit shall have a plug receptacle and circuitry that is compatible with the tape punch output of the system, punching both paper and mylar tapes of dimensions, characteristics and code in accordance with 3.4.10.2. The punching speed of the unit shall be not less than 75 characters per second.

3.4.10.38 Postprocessor. When specified (see 6.2.1), the system shall be furnished with a postprocessor program(s) to be utilized in off-line computer assisted preparation of part programs. The postprocessor shall be provided for converting specified programming language (see 6.2.1) to the form and format required to operate the machine tool and control. The postprocessor shall be written for use on the type computer specified (see 6.2.1). Documentation sufficient to implement and utilize the postprocessor shall be provided for computer compatibility verification and validation at least 30 days prior to delivery of the machining center.

3.4.10.39 Graphics software. When specified (see 6.2.1), graphics software shall be provided which will allow communication between the specified (see 6.2.1) user graphics system and the machine control unit.

3.4.11 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with the machine:

a. A load meter installed on the machine or MCU console that constantly registers the horsepower load (horsepower units or percent of load) on the spindle rotational drive motor.

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b. One set of special wrenches, operating, and repair tools that are normally furnished by the manufacturer.

3.4.12 Optional equipment. The following optional equipment shall be furnished as specified (see 6.2.1):

a. A worklight, conveniently mounted to illuminate the work area, complete with protective shield and on-off switch.

b. An elapsed time indicator of the non-reset type, with digital readout and designed to resist the effects of vibration, voltage fluctuation, temperature change, and humidity. The indicator shall be mounted on the machine in a position visible from the operator's station and be connected to record and accumulate machine operating time to not less than 9999.9 hours.

c. Tools, tool holders, and adapters, as specified.

d. Other optional equipment as specified.

3.5 Sizes and capacities. Unless otherwise specified (see 6.2.1), the sizes and capacities of the machining center shall meet the requirements of table I as applicable for the machine size specified in the contract or order.

3.6 Performance. The machine and numerical control system combination shall be capable of performing as specified herein. In addition, the machine shall be capable of performing drilling, tapping, milling and boring to accomplish the capacities of table I while meeting the accuracies of table II.

3.7 Alignment. The machine and numerical control system shall meet the alignment and accuracy tolerances of table III.

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TABLE I. Sizes and capacities.

CHARACTERISTIC	REQUIREMENTS							
	Size 1	Size 2	Size 3	Size 4	Size 5	Size 6	Size 7	Size 8
Spindle Motor horsepower	5	7	7	10	10	10	15	20
Tape controlled travel, inches:								
x axis	30	30	50	40	50	80	54	54
y axis	15	15	26	20	26	26	28	28
z axis	8	10	10	12	12	12	14	24
Capacity (in SAE 1020 steel):								
Drill, dia., inches	1	1-1/4	2	1-3/4	1-3/4	1-3/4	2	2
Tap, dia., inches	1-1/4	1-1/4	2	1-1/2	1-1/2	1-3/4	2	2
Mill, cubic in. per min.	5	7	7	10	10	10	15	20
Machining head & spindle assembly:								
Spindle speeds, number	12	32	80	32	32	32	32	50
Spindle speed range, RPM	100 to 2000	100 to 2000	45 to 3000	45 to 1500	70 to 3000	70 to 3000	30 to 3000	30 to 3000
Spindle feeds, number,	inf.	inf.	inf.	inf.	inf.	inf.	inf.	inf.
Spindle feed rate range, in. per min.,	0.5 to 90	0.5 to 100	0.2 to 100	0.5 to 200	0.5 to 200	0.5 to 200	0.5 to 200	0.5 to 150
Spindle travel (z axis), in.	8	10	10	12	12	12	14	24
Spindle rapid traverse rate, in. per min.	80	100	200	200	200	200	200	200
Machining head or worktable vertical travel (z axis), in.	8	10	10	15	20	20	20	20
Machining head or worktable vertical rapid traverse rate, in. per min.	80	100	200	200	200	200	200	200
Worktable:								
Work mounting surface, length x width, in.	38 x 16	38 x 16	54 x 26	44 x 20	54 x 26	80 x 26	60 x 28	60 x 28
Load capacity, lbs.	1000	2000	2500	2500	3000	3500	3500	3500
Worktable (saddle or column) table, in.								
x axis	40	40	50	40	50	80	54	72
y axis	20	20	26	20	26	26	28	28
Rapid traverse rate, in. per min.	120	150	200	200	200	200	200	200
Milling feeds, number	inf.	inf.	inf.	inf.	inf.	inf.	inf.	inf.
Milling feed rate range, in. per min.	0.5 to 90	0.5 to 100	0.2 to 100	0.5 to 200	0.5 to 200	0.5 to 200	0.5 to 200	0.5 to 150
Tool Changer:								
Chip to chip time (successive operations two different tools in the ready position), seconds	12	12	12	12	12	12	12	12

NOTE: Machine size and capacities shall be not less than the stated requirements of the size ordered nor greater than the requirements of the next larger size shown in above table. When the largest size and capacity machine is ordered the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the minimum or less to the maximum or greater.

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TABLE II. Performance accuracy.

MACHINING OPERATION	ACCURACY (Permissible tolerance)
Drilling:	
Depth of hole -	±0.001"
Tapping:	
Threaded holes shall be Class III fit in accordance with FED-STD-H28	
Boring and Counterboring:	
Hole location, based on centerline distance of bored holes, 2 holes, 18" apart	+0.002"
Hole size, maximum spread, hole to hole	±0.002"
Depth of counterbore -	±0.001"
Milling (straight-line):	
Squareness of sides	0.002"/FT
Parallelism of opposing sides	0.002"/FT
Flatness	+0.002"/3 FT
Angularity of ramp cuts	±0.002"/FT
Corner location, deviation from programmed points	±0.001"
Contouring (circle):	
Roundness	
Diameters measured at 30 degree intervals	+0.002"
Radii measured at 30 degree intervals	±0.002"
Surface finish, maximum roughness 64 RMS	
Angularity	
Difference between programmed and actual points of tangency of circles to straight surfaces	0.002"
Contouring (straight-line):	
Squareness of sides	0.002"/FT
Angularity of sides	0.002"/FT

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TABLE III. Alignment and accuracy.

COMPONENT CHARACTERISTIC	PERMISSIBLE TOLERANCE
Worktable:	
Flatness of work mounting surface	0.001"/FT
Squareness of work mounting surface to spindle axis, in any 6" sweep radius	0.002"/FT
Motion in the x axis, rise and fall with respect to the spindle	0.001"/FT
Motion in the y axis, rise and fall with respect to the spindle	0.001"/FT
Parallelism of T-slots with x axis movement	0.0005"/FT
Squareness of T-slots with y axis movement (slide travel squareness)	0.0005"/FT
Spindle:	
Axial runout	
Radial runout:	
1" from spindle nose	0.0005"TIR
6" from spindle nose	0.0015" TIR
Squareness of spindle movement to the work mounting surface of the worktable per 6" of travel	0.0005"
Numerical control system:	
x & y axes positioning and repeatability accuracy:	
absolute positioning	1/ ±0.001"/FT
position repeatability	±0.0005"
z axis positioning and repeatability accuracy:	
absolute positioning	±0.001"/FT
positioning repeatability	±0.0005"
a axis (when furnished) positioning and repeatability accuracy:	
absolute positioning	±10 sec./FT
positioning repeatability	±5 sec.
Axis overshoot:	
20% of maximum feed rate	0.003"
Maximum feed rate	0.005"

1/ Not to exceed 0.025" in overall travel.

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TABLE IV. - Diskette addressing.

Sector	*	Sector*	Byte	Use
0-25	0	All	All	Unused
26-29	1	1-4	0-1	File start sector (Binary)
			2-31	30 character ASC11 program name to be defined by user
			32-63	Same as 0-31 above for next Part program (file)
			64-95	Same as 32-63 above for next part program
			96-127	Same as 32-63 above for next part program
		5-26	All	APT CL records for NC part programs
30-1897	2-73	All	All	APT CL records for NC part programs
1898-1975	74-76	All	All	Unused

* This numbering corresponds to IBM System 3740 physical format.

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3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461.

3.9 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubrication application
Servicing interval
Type of lubricant
Viscosity

3.10 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. Data shall be furnished as specified (see 6.3).

3.12 Workmanship. Workmanship of the machine and accessories shall be equal to that of the manufacturer's current commercial machines of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of the inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection. When first article approval is required under 3.1, first article inspection shall be performed. Unless otherwise specified (see 6.2.1), first article inspection shall be comprised of the examination in 4.4 and all tests in 4.5. Failure to pass the examination or any test shall be cause for disapproval of the first article.

4.3 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5 through 4.5.5, and the inspection in 4.6. Failure of the item to pass any examination or test shall be cause for rejection.

4.4 Examination. The machine and equipment shall be examined for compliance with the requirements in 3.2 through 3.5 and 3.9 through 3.12.

4.5 Tests. The machine shall be subjected to the following tests. All instruments, tapes, materials, and tools required to perform and evaluate these tests shall be furnished by the supplier. The measuring instruments shall have evidence of calibration traceable to the National Bureau of Standards. The numerical control test tapes and computer printouts used shall become the property of the Government.

4.5.1 Operational test. The machine and its numerical control system shall be operated in accordance with the manufacturer's standard operating test procedure for warm-up and run-off checks. During the warm-up period, proper operation of all manual controls, motors, adjustment mechanisms and accessories shall be verified. After warm-up, the machine shall be cycled continuously under numerical tape control for a period of not less than 4 hours. This operation shall include tool change, spindle speed and feed rate changes that include the highest, intermediate, and lowest settings of each range, rapid traverse of all slides, simultaneous movement of slides and automatic feed cycles as applicable. The numerical control system shall be further tested to verify proper operation of MDI, program edit, and system diagnostics. Should a malfunction occur, it shall be corrected and the operational test repeated until a full 4 hours of running time is completed without failure.

4.5.2 Alignment and accuracy test. The machine and its numerical control system shall be tested for conformance with the alignment and accuracy requirements of table III.

4.5.2.1 Axis positioning and overshoot tests. Starting at positions other than the extremes of axis travel, two identical movements of each controllable linear axis shall be programmed and executed in each direction. Programmed span shall be not less than 7 inches, and each digit of starting and end points shall be other than zero with respect to reference zero. Feed rate of axis shall be 2 inches per minute. Absolute positioning and positioning repeatability errors shall not exceed the permissible tolerances of table III. Similar movements of each linear axis shall be

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programmed at varied feed rates to test for slide overshoot at accelerated feed rates without programmed deceleration. One pass shall be programmed at feed rates nearest to 10, 20 and 100 percent of maximum linear feed rate. Overshoot shall not exceed 0.003 inch at any feed rate up to 20 percent of the maximum. Maximum overshoot shall not exceed 0.005 inch. Stops shall be programmed appropriately for inspection of control positioning accuracy. Positioning tests may be part of the same tape program as 4.5.1.

4.5.3 Performance accuracy test. While using one or more test setups devised by the supplier, the machine shall be tested to verify its ability to perform machining operations under numerical control for meeting the accuracy tolerances of table II. The test piece(s) may be, at the supplier's option, steel, cast iron or aluminum and the tool(s), spindle speed and feed rate for each test shall be selected by the supplier as the most suitable for the particular operation being performed. The test piece(s) shall present a work surface normal to the tool point and approximately centered with reference to the work area in the xy plane. In milling and contouring test, successive cuts may be made inside the outlines of previous cuts and roughing cuts may be made in the various cuts with excess metal removed as desired to make measurement convenient.

4.5.3.1 Tapping test. Pilot drill and tap at programmed locations, four holes each with finished threads to accept 8-32 and 3/4-10 screws. Finished holes shall accept standard 8-32 or 3/4-10 screws freely and threads shall meet Class 3 standards for internal threads as set forth in FED-STD-H28.

4.5.3.2 Boring test. Pilot drill, semi-finish bore, and finish bore not less than 10 holes in a selected pattern covering an area not less than 24 inches in diameter. Four of the holes shall be spaced on 18 inch centers. Different tools shall be used for semi-finish and finish boring operations. All holes may be pilot drilled in sequence, but semi-finish and finish boring of each hole shall be completed with necessary tool changes before beginning boring operations of successive holes. Each address shall be to the fourth decimal place with digits other than zero. Finished hole size shall range from 1.125 to 2.750 inch. The resulting hole location and hole size errors shall not exceed the permissible tolerance of table II for boring.

4.5.3.3 Straight-line milling test. The straight line milling test shall be a programmed square or rectangle, not less than 12 inches per side with sides parallel to the travel along x and y axes. Machines shall make a plunge cut along one longitudinal side cut. Plunge angle shall be not less than 2 degrees. The resulting cuts shall be checked for dimensional accuracy, squareness, parallelism, flatness, angularity, and corner locations to verify the accuracy required in table II for straight-line milling.

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4.5.3.4 Circle contouring test. A circle shall be programmed and milled within the outline of the square cut required in 4.5.3.3. The cut shall be made at constant z axis depth of one-half inch and by using the shortest programming sequence within the capability of the MCU. The resulting cut shall be checked for dimensional accuracy, roundness, surface finish, and angularity to verify the accuracy required in table II for circle contouring.

4.5.3.5 Straight-line contouring test. A square or an equilateral triangle shall be programmed and milled within the circle cut and described by 4.5.3.4 at constant depth of one-half inch. The reference line for one side of the square or triangle shall be 10 to 20 angular degrees canted to the x axis. The resulting cut shall be checked for dimensional accuracy, squareness and angularity to verify the accuracy required in table II for straight-line contouring.

4.5.4 Maximum horsepower test. Using a workpiece of SAE 1020 steel, a straight cut of not less than 12 inches in length shall be made with a face mill or shell end mill. Diameter of the cutter and spindle speed shall be determined by the supplier to be suitable for making the cut with specified chip load and peripheral cutter speeds. The depth of the cut shall be as required to load the spindle motor to its maximum rated horsepower. Chip load shall be 0.010 $\pm$0.002 and peripheral cutter speed shall not exceed 300 feet per minute. The cutting action shall be smooth and even and the finished workpiece shall show no evidence of tool chatter. In addition, there shall be no evidence of overheating of the spindle drive motor or the applicable feed motor.

4.5.5 Test program verification. Unless otherwise specified (see 6.2.1),, while using one or more test set-ups devised by the supplier, the machining test program (tape) required under postprocessor validation (see 3.4.10.38), shall be run on the machining center to verify its ability to receive and perform all program functions without malfunction or error. The test piece(s) shall be either steel, cast iron, or aluminum, while the tooling shall be selected by the supplier as the most suitable for the particular operation being performed.

4.5.6 Electromagnetic interference control tests. Unless otherwise specified (see 6.2.1), equipment requiring electromagnetic interference control testing shall be tested for compliance with 3.8.

4.6 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with ASTM-D-3951. When specified (see 6.2.1), level A packaging, level A or Level B packing and special marking shall be accomplished in accordance with MIL-M-18058.

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6. NOTES

6.1 Intended use. The numerically controlled machines covered by this specification are intended for use in any production shop where drilling, boring, tapping, reaming, and milling operations by numerical control of the machine are required. The machine may be used to produce prototype parts from a taped processor to produce production items repetitively from a proven process, to accuracies within the capabilities of the machine.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchasers should specify their requirements in the procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.
b. Style and size required (see 1.2).
c. First article approval, if required (see 3.1).
d. Specify means of accomplishing z axis movement, if different (see 3.2).
e. Specify additional safety and health requirements (see 3.2.2 and 6.5).
f. Painting, if different (see 3.3.5).
g. Dials, if different (see 3.3.7).
h. Hardened and ground guideways, if required (see 3.4.3).
i. Specify the number and configuration of T-slots, if different (see 3.4.4).
j. Sub-plate and desired characteristics, if required (see 3.4.4).
k. Specify special characteristics for spindle noses, and bores if required (see 3.4.5.1).
l. Coolant system, if different (see 3.4.7).
m. Specify noise level and enclosure characteristics for stand-alone hydraulic units, if required (see 3.4.8).
n. Electrical system, electrical supply, and voltage for initial wiring, if different (see 3.4.9).
o. Electric motor enclosures, if different (see 3.4.9.1).
p. Specify EIA-RS-494 Binary Cutter Location (BCL) data, if required (see 3.4.10).
q. Specify the program selectable (degree) positions required (72 - 360 - 360,000, etc.) and whether the a axis is required to function simultaneously or independently of the x and y axes of motion, if a rotary table is required (see 3.4.10.1.1).
r. Specify type of rotary table (see 3.4.10.1.1).
s. Specify EIA-RS-244 in lieu of EIA-RS-358 (see 3.4.10.2.1).
t. Specify type of take-up spools, if not standard (see 3.4.10.2.1).
u. Specify BCL Exchange Input, when required (see 3.4.10.2.2).

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v. Data display, if different (see 3.4.10.3).
w. Specify memory type, if different (see 3.4.10.9).
x. Specify part program storage capacity, if different (see 3.4.10.9).
y. Specify the hours of standby battery power required, if different (see 3.4.10.10).
z. Specify additional NC features, if different (see 3.4.10.26).
aa. Operator control panel functions, if different (see 3.4.10.27).
bb. Pendant control panel, if required (see 3.4.10.27.1).
cc. Specify pendant control functions, if different (see 3.4.10.28).
dd. Specify one inch punched tape or eight inch flexible disk as MCU programs input and storage media (3.4.10.31.1).
ee. Specify fault conditions required, if different (see 3.4.10.32).
ff. Specify type of equipment interface, if different (see 3.4.10.33).
gg. Specify auxiliary and miscellaneous functions, if different (see 3.4.10.35).
hh. Tape punch unit, if required (see 3.4.10.37).
ii. Postprocessor, specify computer name and model, form and format (see 3.4.10.38).
jj. Graphics software (specify graphics system name and model, form and format) if required (see 3.4.10.39).
kk. Specify standard equipment, if different (see 3.4.11).
ll. Optional equipment, if required; fully describe (see 3.4.12).
mm. Specify size and capacities, if different (see 3.5).
nn. Electromagnetic interference control, if required (see 3.8).
oo. Lubrication chart, if different (see 3.9).
pp. Nameplate, if different (see 3.10).
qq. First article inspection, if different (see 4.2).
rr. Quality conformance inspection, if different (see 4.3).
ss. Test program verification, if different (see 4.5.5).
tt. Electromagnetic interference control test requirements, if different (see 4.5.6).
uu. Preservation, packaging and marking, if different (see 5.1).

6.3 Contract data requirements. All data requirements such as parts lists, wiring diagrams, operating manuals, maintenance manuals, alignment and test results shall be requested on a DD Form 1423, Contract Data Requirements List, and shall be furnished in accordance with each specified DD Form 1664, Data Item Description, as shown on the DD Form 1423.

6.4 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.5 Safety and health requirements. Paragraph 3.2.2 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat,

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etc. Since OSHA limits the total level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the requesting activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. If specific point-of-operation guarding is required, as in most cases, the guard configuration is dependent on the size and configuration of workpieces. The above, and any other additional safety and health requirements, should be specified in detail (under 6.2.1.e).

6.6 Mercury prohibited. When mercury or its compounds are to be prohibited from use in the machine and its components and accessories, the procurement documents should include a statement to that effect.

6.7 Asbestos prohibited. When the use of either asbestos or its fibers in the machine accessories is to be prohibited, procurement documents should include a statement to the effect.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review Activities:
Army - AL
Navy - SH, AS
Air Force - 99
DLA - GS

Preparing Activity:
DLA - IP

Project Number:
3408-0016

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-80255A	Machining Centers, Vertical, Single Spindle, Computer Numerical Control (CNC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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